Web Developer Intern (MERN Stack) – Technical Assignment

Weather Forecast & Advisory Tool for Farmers

Overview

Build a simple web application using the MERN stack that fetches real-time forecast data from the **OpenWeatherMap API (free tier)** and converts it into **useful, farmer-friendly advisories**.

Assignment Details

1. Build a MERN Web Application

- Use React.js for frontend
- Node.js + Express for backend
- MongoDB or any other DB for storing user search history (optional but recommended)

2. Weather Data Fetching

- Input: location name
- Fetch following from OpenWeatherMap:
  - Temperature
  - Humidity
  - Rain probability (POP)
  - Wind speed
  - 5-day forecast (3-hour blocks)

3. Farmer Advisory Logic

After fetching the weather, generate simple, rule-based advisories.

Examples (you may define your own):

- **Rain Probability > 60%**
  Avoid irrigation and pesticide spraying today.
- **High Temperature (> 35°C)**
  Increase irrigation frequency for heat-sensitive crops.
- **Wind Speed > 15 km/h**
  Do not spray pesticides due to drift risk.
- **High Humidity (> 80%)**
  Possible fungal infection; monitor your crop.
- **Good Spraying Window**
  If wind < 10 km/h and no rain expected in next 6 hours.

Use at least **4–5 advisory rules**.

4. Display UI

Your UI should show:

- Basic weather metrics
- Graph or small chart for temperature/rain trend
- Generated advisories in an eye catchy text box.

5. Extra (Optional)

- Save last 5 searched locations in MongoDB
- Download advisory as PDF in presentable manner.

Evaluation Criteria

- Code clarity and folder structure
- Code documentation
- Correct API integration
- Advisory logic quality
- UI clarity and usability
- Bonus: any creative, meaningful addition for farmers

Submission Requirements

- GitHub repository with full source code
- README with setup instructions
- Short demo video (2–3 min) showing:
  - Weather input
  - Weather data display
  - Advisory generation

Deadline

Friday, 12th Dec 2025 by 12:00 PM.